

Focus on Internetworking	Initially did not account for internetworking , assuming each country would use a national OSI-based network.	Internetworking was a core design goal from the beginning.
Protocol Dependency	Protocol-independent , no specific protocols defined; focuses on structure and services.	Protocol-specific , designed around key protocols like IP, TCP, and UDP.

Question and Answers

Short Answer Questions

1. Define a computer network (Jan. 2025)

Ans. A computer network is a collection of interconnected computers and devices designed to communicate and share resources such as data, files, software applications, and hardware components.

2. List any four uses of computer networks:

Ans. Computer networks are used for business operations like inventory and payroll, for accessing the Internet at home, enabling mobile users to stay connected while on the move, and for social applications like instant messaging and online collaboration.

3. Expand VPN and RFID:

Ans. VPN stands for Virtual Private Network, which enables secure remote access to a network. RFID stands for Radio Frequency Identification, used to wirelessly identify and track objects using tags and readers.

4. Expand GPS and NFC:

Ans. GPS stands for Global Positioning System, a satellite-based navigation system. NFC stands for Near Field Communication, a technology that enables short-range communication between compatible devices.

5. **Expand DMCA and CAPTCHA:**

Ans. DMCA stands for Digital Millennium Copyright Act, a U.S. law protecting digital content rights. CAPTCHA stands for Completely Automated Public Turing test to tell Computers and Humans Apart, used to prevent bots from accessing websites.

6. **Expand WiMAX and NSP:**

Ans. WiMAX stands for Worldwide Interoperability for Microwave Access, a wireless communication standard. NSP stands for Network Service Provider, which offers internet services to customers.

7. **Expand ARPANET and ICMP:**

Ans. ARPANET stands for Advanced Research Projects Agency Network, the first packet-switching network. ICMP stands for Internet Control Message Protocol, used for error reporting and diagnostics in IP networks.

8. **Expand SMTP and RTP:**

Ans. SMTP stands for Simple Mail Transfer Protocol, used for sending emails. RTP stands for Real-time Transport Protocol used for delivering audio and video over IP networks.

9. **Expand HTTP and TELNET:**

Ans. HTTP stands for HyperText Transfer Protocol, used for web communication. TELNET stands for TELEcommunication NETwork, a protocol for remote login to other computers.

10. **Expand FTP and DNS:**

Ans. FTP stands for File Transfer Protocol, used to transfer files over a network. DNS stands for Domain Name System, which translates domain names into IP addresses.

11. **Expand TCP and UDP:**

Ans. TCP stands for Transmission Control Protocol, which ensures reliable data transmission. UDP stands for User Datagram Protocol, a faster but unreliable protocol used in real-time applications.

12. **Expand DSL and SONET:**

Ans. DSL stands for Digital Subscriber Line, a technology for

high-speed internet over telephone lines. SONET stands for Synchronous Optical Network, a standard for optical communication.

13. **Expand LAN, MAN, and WAN:**

Ans. LAN is Local Area Network, MAN is Metropolitan Area Network, and WAN is Wide Area Network. They vary based on the geographic area covered—from a single building to across countries.

14. **What is the peer-to-peer model in computer networks?**

Ans. In the peer-to-peer model, each device can act as both a client and a server, sharing resources directly with others without centralized control.

15. **Define the client-server model with an example:**

Ans. In the client-server model, servers provide resources or services while clients request them. For example, a web browser (client) requesting pages from a web server.

16. **List any four forms of e-commerce:**

Ans. The different forms of e-commerce include Business-to-Business (B2B), Business-to-Consumer (B2C), Consumer-to-Consumer (C2C), and Mobile Commerce (M-commerce).

17. **What is profiling in the context of internet usage?**

Ans. Profiling refers to the collection and analysis of user behavior and preferences based on their online activities, often used for targeted advertisements.

18. **What is phishing in online security?**

Ans. Phishing is a cybercrime where attackers impersonate legitimate entities through email or websites to steal sensitive information like passwords or credit card details.

19. **Mention the two types of transmission technologies used in networks:**

Ans. The two main transmission technologies are broadcast transmission, where data is sent to all nodes, and point-to-point transmission, where data is sent from one source to a specific destination.

26. Define unicasting and broadcasting:

- 20. Ans.** Unicasting is communication from one sender to one receiver. Broadcasting is communication where data is sent from one sender to all devices on the network.

21. Differentiate between static and dynamic allocation methods:

- Ans.** Static allocation pre-assigns resources, often inefficient in varying conditions. Dynamic allocation assigns resources based on demand, improving flexibility and efficiency.

22. What is a communication subnet?

- Ans.** A communication subnet refers to the part of the network responsible for routing data, consisting of transmission lines and switching elements like routers.

23. List the main components of a communication subnet:

- Ans.** The two main components are transmission lines, which carry data between devices, and switching elements such as routers, which direct data to the appropriate destination.

24. What is the difference between routing and forwarding?

- Ans.** Routing is the process of determining the best path for data packets. Forwarding is the actual movement of packets from one router to another based on routing decisions.

25. Define addressing and flow control in a network:

- Ans.** Addressing identifies the source and destination devices in a network. Flow control manages the pace of data transmission to prevent buffer overflow or data loss.

26. Differentiate between connection-oriented and connectionless services:

- Ans.** Connection-oriented service establishes a dedicated path before communication (e.g., TCP). Connectionless service sends data without prior setup (e.g., UDP), suitable for fast, low-latency communication.

27. What is the difference between store-and-forward switching and cut-through switching?

- Ans.** In store-and-forward switching, the entire frame is received and checked for errors before forwarding. In cut-through switching,

forwarding begins as soon as the destination address is read.

28. List any four service primitives in networking:

- Ans.** Common service primitives include LISTEN, CONNECT, SEND, and RECEIVE. These operations enable communication setup, data exchange, and disconnection between entities.

29. Define service and protocol with examples:

- Ans.** A service is a set of operations provided by one layer to the layer above, like reliable data transfer. A protocol is a set of rules that govern data exchange, such as HTTP or FTP.

30. Write any two differences between the OSI and TCP/IP reference models:

- Ans.** The OSI model has 7 layers, while TCP/IP has 4. OSI separates services, interfaces, and protocols clearly; TCP/IP combines them, making implementation more practical but less modular.

31. What is multicasting in computer networks?

- Ans.** Multicasting is a transmission method where data is sent from one source to a selected group of recipients, rather than to all devices (broadcast) or a single recipient (unicast).

32. What is the purpose of the session layer in the OSI model?

- Ans.** The session layer establishes, manages, and terminates sessions between applications, ensuring synchronized and organized communication between systems.

33. What is meant by flow control in networking?

- Ans.** Flow control is a technique used to prevent the sender from overwhelming the receiver with too much data too quickly, ensuring smooth and error-free data transfer.

34. What is meant by error control in data communication?

- Ans.** Error control involves detecting and correcting errors in transmitted data using acknowledgments, retransmissions, and error-detection techniques like checksums or CRC.

35. What is an internetwork?

- Ans.** An internetwork is a collection of interconnected but different types of networks (e.g., LAN, WAN, MAN) that function as a single network, with the Internet being the best example.

Long Answer Questions**1. List and explain the major uses of computer networks.**

Ans. Computer networks have become a fundamental component of modern communication and information systems. Their uses span across various domains, providing services that enhance productivity, convenience, and global connectivity. The major uses of computer networks are:

2. Business Applications:

Computer networks are extensively used in businesses for communication, resource sharing, and automation of tasks. They support activities such as email communication, online transactions, database access, and inventory control. Networks allow employees to collaborate in real time using shared tools and documents, which leads to improved efficiency and reduced costs.

3. Home Applications:

At home, networks are used for internet access, online education, entertainment (such as video and music streaming), social networking, and online shopping. Multiple devices like smartphones, computers, and smart TVs can be interconnected to share resources and internet connectivity.

4. Mobile Users:

Mobile users access the internet or a private network while in motion, using wireless technologies such as Wi-Fi, 4G/5G, or mobile hotspots. This allows continuous connectivity to cloud services, emails, and web applications irrespective of the user's location.

Social Issues:

Networks have reshaped how people interact socially and professionally. They enable virtual meetings, social media platforms, and e-governance. However, they also raise concerns like digital divide, cybercrime, privacy breaches, and addiction to online platforms.

2. Write a short note on the following applications of computer networks:

Ans. **a) Business Applications:** (Jan. 2025)

In businesses, computer networks are used extensively to support operational and administrative functions. Common business applications include email communication, access to centralized databases, collaborative tools like file sharing, video conferencing, and electronic fund transfers. Intranet portals are used within organizations for employee access to information, policies, and services. Remote access through Virtual Private Networks (VPNs) enables employees to work from anywhere while securely connecting to the company's internal systems. Networks thus improve communication, boost productivity, and reduce operational costs.

b) Home Applications:

In the home environment, computer networks provide connectivity for multiple devices such as laptops, desktops, smartphones, smart TVs, and tablets. These networks are primarily used for accessing the internet, online shopping, entertainment (e.g., video and music streaming), online gaming, social networking, and remote learning. Home networks are typically established using wireless routers that connect devices to the internet. Additionally, smart home devices like security systems and automation tools are also integrated into home networks for convenience and control.

3. Write a short note on the following

a) Mobile Users: b) Social Issues:

Ans. a) Mobile Users:

Mobile users refer to individuals who use portable devices like smartphones, laptops, and tablets to connect to networks while on the move. With the support of wireless technologies such as Wi-Fi, 4G, and 5G, users can remain connected to the internet or enterprise networks from virtually any location. For example, sales personnel may access inventory databases remotely or employees may attend virtual meetings from remote

locations. The ability to stay connected while mobile enhances productivity and responsiveness.

b) Social Issues:

While computer networks have enabled numerous advancements, they also give rise to various social concerns. These include issues such as data privacy, identity theft, online frauds, digital addiction, and cyberbullying. The digital divide, where certain populations have less access to technology than others, is another growing concern. Additionally, the misuse of online platforms for spreading misinformation and hate speech has become a major challenge in the digital age.

4. Explain the following types of computer networks with examples

(a) Local Area Network (LAN):

(b) Metropolitan Area Network (MAN):

Ans. a. Local Area Network (LAN)

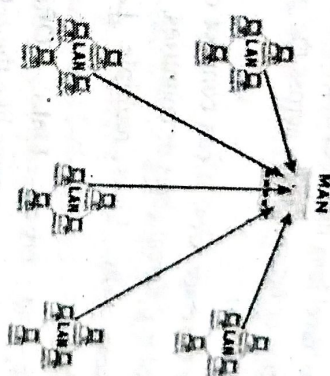


A Local Area Network (LAN) is a computer network that connects devices within a limited geographical area such as a single room, building, or campus. It enables users to share resources like files, printers, software applications, and internet connectivity among connected devices. LANs are known for offering high-speed data transfer rates, low latency, and reliable communication.

They are typically owned, controlled, and maintained by a single organization or individual, which ensures better security and ease of management. LANs may use wired connections (such as Ethernet cables) or wireless technologies (like Wi-Fi). A classic example of a LAN is a college computer laboratory where all computers are connected to a central server, allowing students to access shared files and applications. Other examples include office networks and home Wi-Fi networks.

b.

Metropolitan Area Network (MAN):



A Metropolitan Area Network (MAN) spans a broader geographical area than a LAN, usually covering a town, city, or a group of nearby buildings. It is designed to connect multiple LANs within the specified region to form a larger and more powerful network. MANs are often owned and operated by telecommunications companies, municipal bodies, or large organizations, and they may use fiber-optic cables, high-speed switches, and routers to interconnect the LANs. MANs are typically used to provide broadband internet access, support video conferencing, and enable data exchange between branch offices within the same city. A practical example of a MAN is the network that connects all branches of a university located across a city, allowing seamless communication, centralized data management, and resource sharing among campuses.

5. What is a Wide Area Network (WAN)? Explain its structure, features, and uses with an example.

Ans. A Wide Area Network (WAN) is a type of computer network that covers a very large geographical area, often spanning cities, countries, or even continents. It is used to connect computers and smaller networks (like LANs and MANs) that are far apart from each other.

Structure of WAN:

A WAN is made up of many smaller networks, such as Local Area Networks (LANs) and Metropolitan Area Networks (MANs), which are connected using routers, leased telephone

lines, fiber-optic cables, or satellite links. These components allow data to travel long distances and connect different parts of an organization or network users across the globe.

Features of WAN:

- ◇ **Large Coverage Area:** It can connect offices or branches located in different cities or countries.
- ◇ **Interconnectivity:** Links multiple LANs and MANs together.
- ◇ **Public and Private Links:** Often uses public telecommunication systems like the telephone network or the Internet, along with private leased lines.
- ◇ **Lower Speed than LANs:** Because of the long distances and many connections involved, WANs usually operate at slower speeds compared to LANs.
- ◇ **High Setup Cost:** Building and maintaining a WAN is more expensive due to the infrastructure and equipment involved.

Applications/Uses of WAN:

- ◇ Used by multinational companies to connect their regional offices and share data.
- ◇ Used by government departments for centralized services like tax systems or national ID databases.
- ◇ Supports global communication, including email, websites, video conferencing, and cloud services.

6. What is an internetwork? Explain with an example.

Ans.

An internetwork (often referred to as an internet with a small "i") is a network that is formed by interconnecting two or more separate computer networks, using devices such as routers and gateways. These networks may differ in terms of technology, architecture, or communication protocols, but the internetwork allows them to function together as a single, unified system.

Benefits of Internetworking:

- ◇ Enables global communication and collaboration.
- ◇ Allows sharing of resources (such as printers, files, and databases) across different networks.

- ◇ Supports scalability, meaning new networks or devices can be added easily.
- ◇ Offers fault tolerance and redundancy through multiple paths for data.

Example

The most well-known and widely used internetwork is the Internet. It is the largest internetwork in the world, connecting millions of networks—ranging from home networks and small business networks to government and corporate systems. The Internet allows users to access websites, send emails, use cloud applications, and connect with others worldwide.

7. Explain the concept of protocol hierarchies in layered network architecture (Jan. 2025)

Ans.

Protocol hierarchies refer to the structured and layered arrangement of protocols in a computer network, where each layer is responsible for a specific set of functions. This concept is fundamental to the design of modern network architectures, such as the OSI (Open Systems Interconnection) model and the TCP/IP model.

In a protocol hierarchy, each layer performs a well-defined role in the process of data communication. The key idea is that each layer provides services to the layer directly above it and receives services from the layer directly below it. This design approach is known as the layered architecture.

How Protocol Hierarchies Work:

Each layer in the hierarchy communicates with its peer layer on the other machine using a specific protocol, which is a set of rules that defines how data is formatted, transmitted, and received. However, the actual data transfer is handled by the lower layers.

For example:

- ◇ The application layer on one computer sends a message using a protocol like HTTP.
- ◇ This message is passed down to the transport layer, which may use TCP to ensure reliability.

- ◇ The network layer then decides the routing path using IP.
- ◇ The message finally travels through the physical medium after being processed by the data link and physical layers.
- In the **client-server model**, the server is a central entity that provides services such as file storage, printing, or database access. Clients send requests and receive responses from the server. This model is easy to manage and secure but depends heavily on the server's availability.
- In the **peer-to-peer (P2P) model**, all devices (peers) are equal and can act as both clients and servers. Resources are distributed among peers. It is more scalable and resilient but harder to secure and manage. Examples include file-sharing applications like BitTorrent.

8. **What is the client-server model? How does it differ from the peer-to-peer model?:**

Ans. In the **client-server model**, the server is a central entity that provides services such as file storage, printing, or database access. Clients send requests and receive responses from the server. This model is easy to manage and secure but depends heavily on the server's availability.

In the **peer-to-peer (P2P) model**, all devices (peers) are equal and can act as both clients and servers. Resources are distributed among peers. It is more scalable and resilient but harder to secure and manage. Examples include file-sharing applications like BitTorrent.

9. **Discuss the design issues that must be addressed in the layered approach of network models:**

Ans. In the layered architecture of computer networks, several key design issues must be addressed:

- ◇ **Addressing:** Each device must have a unique identifier to facilitate communication.
- ◇ **Error Control:** Ensures data integrity through mechanisms like checksums and acknowledgments.
- ◇ **Flow Control:** Prevents the sender from overwhelming the receiver with excessive data.

- ◇ **Sequencing:** Ensures data packets are received in the correct order.
- ◇ **Multiplexing:** Allows multiple users to share the same communication channel efficiently. These issues are handled independently at each layer, contributing to the reliable and efficient functioning of the network.

10. **Differentiate between connection-oriented and connectionless services with suitable examples:** (Jan. 2025)

Ans. **Connection-oriented services** require the establishment of a dedicated connection between the sender and receiver before data can be transmitted. They ensure reliable delivery, sequencing, and error control. An example is TCP (Transmission Control Protocol), which is used for applications like file transfer and email.

In contrast, **connectionless services** do not establish a prior connection and send data packets independently. There is no guarantee of delivery or order, but they are faster and require fewer resources. UDP (User Datagram Protocol) is an example and is used in applications like video streaming and online gaming.

11. **Explain the concept of service primitives with suitable examples:**

Ans. **Service primitives** are the basic operations or functions provided by a network layer to the layer above it, allowing processes to initiate, maintain, use, and terminate communication. These primitives form the interface between adjacent layers in a layered network architecture.

In the context of connection-oriented communication, a typical set of service primitives includes:

- ◇ **LISTEN:** Used by the server to wait passively for incoming connection requests. The server remains blocked until a request arrives.
- ◇ **Example:** A web server waits for incoming HTTP requests.
- ◇ **CONNECT:** Initiated by the client to request a connection with a server. If successful, a reliable communication path is established.

- ◇ **Example:** A browser connects to a web server to download a webpage.
 - ◇ **SEND:** Used to transmit data over an established connection to the connected peer.
 - ◇ **Example:** The client sends a login request to the server.
 - ◇ **RECEIVE:** Used to receive data sent by the peer. The process may block until data is available.
 - ◇ **Example:** The server receives and processes the client's login information.
 - ◇ **DISCONNECT:** Terminates the established connection after the data exchange is complete.
 - ◇ **Example:** The client logs out and disconnects from the server.
- In **connectionless communication**, fewer primitives may be used since no setup or teardown is required. The main primitives are typically:
- ◇ **SEND:** Send a message (datagram) without prior connection.
 - ◇ **RECEIVE:** Receive a datagram from any source.

12. Describe the seven layers of the OSI reference model with their functions: (Jan. 2025)

Ans. The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes the functions of a telecommunication or computing system into **seven distinct layers**, each with specific responsibilities. These layers help different network devices and software communicate effectively.

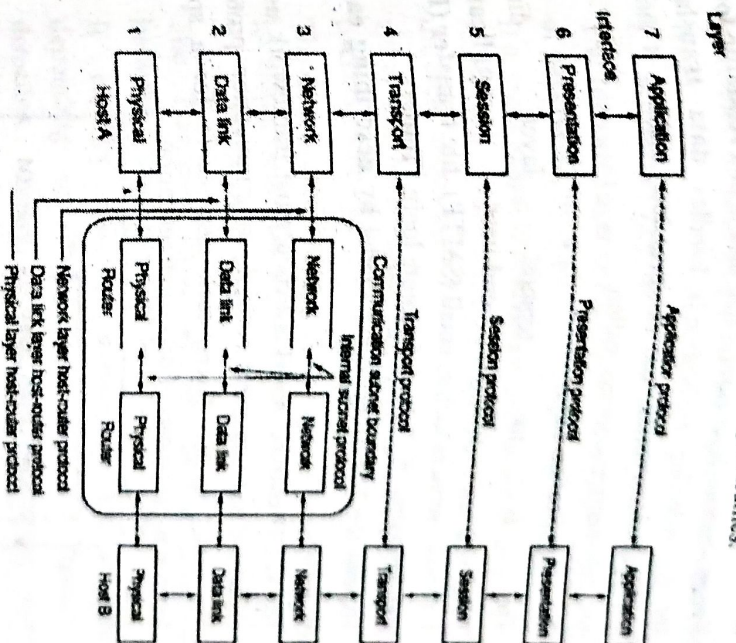
1. Physical Layer

This layer is responsible for the **transmission of raw bits** over a physical medium such as cables, switches, or radio signals. It defines hardware specifications like voltage levels, timing, connector types, data rates, and transmission modes.

2. Data Link Layer

The data link layer ensures **reliable communication** between

two directly connected nodes. It handles **framing, error detection/correction, and flow control**, converting raw bits from the physical layer into meaningful frames.



3. Network Layer

This layer is responsible for **routing and forwarding packets** across networks. It provides **logical addressing** (e.g., IP addresses) and selects the best path to deliver data from source to destination.

4. Transport Layer

The transport layer provides **end-to-end communication** between applications on different hosts. It ensures **reliable data delivery**, performs **error recovery**, and manages **flow control** (e.g., using TCP or UDP).

5. Session Layer

This layer establishes, manages, and terminates sessions or dialogs between applications. It controls the dialog flow,

synchronization, and recovery, ensuring consistent and organized data exchange.

6. Presentation Layer

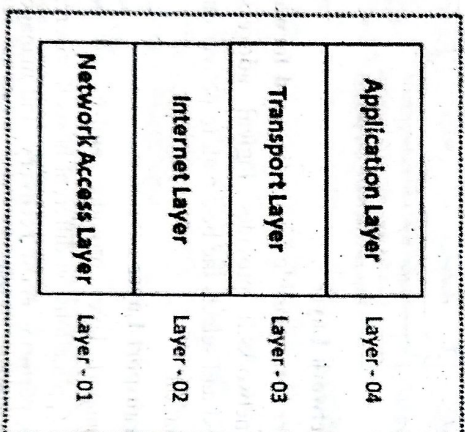
The presentation layer ensures that data is in a readable format for the receiving application. It handles data translation, encryption/decryption, and compression, making it suitable for cross-platform communication.

7. Application Layer

This is the topmost layer, directly interacting with end-user applications. It provides services such as email (SMTP), file transfer (FTP), web browsing (HTTP), and remote login (Telnet).

13. Explain the TCP/IP reference model by describing each of its layers.

Ans. The TCP/IP reference model is a conceptual framework used to understand and implement network communication protocols. The model has four layers, each responsible for a specific function in the process of data communication.



1. Network Access Layer (also called Link Layer):

The Network Access Layer is the lowest layer in the TCP/IP model. It includes all the hardware and software mechanisms

required to place data onto the physical network medium. Its main functions include framing data into packets or frames, assigning physical (MAC) addresses, and performing error detection on the physical link.

Examples of technologies at this layer include Ethernet and Wi-Fi.

2. Internet Layer:

The Internet Layer is responsible for the routing and addressing of packets across multiple interconnected networks.

It assigns logical IP addresses to devices and ensures that packets reach their destination, even if the source and destination are on different networks.

Other functions include fragmentation and reassembly of large packets to suit the requirements of the physical network.

The core protocol used at this layer is the Internet Protocol (IP), including both IPv4 and IPv6.

3. Transport Layer:

The Transport Layer provides end-to-end communication between applications running on different hosts.

It ensures data is delivered reliably or unreliably, depending on the protocol used, and manages error detection, flow control, and congestion avoidance.

The two main protocols at this layer are:

- ◇ TCP (Transmission Control Protocol), which is connection-oriented and reliable, ensuring data arrives correctly and in order.
- ◇ UDP (User Datagram Protocol), which is connectionless and faster, but does not guarantee delivery or order.

4. Application Layer:

The Application Layer is the topmost layer in the TCP/IP model and serves as the interface between user applications and the network.

It provides various network services like file transfer, email, web access, and remote login. Functions at this layer include

user-level interaction, data encoding, encryption, and session management when needed.

Common protocols include HTTP/HTTPS for web access, FTP for file transfers, and SMTP/POP3/IMAP for email services.

13. Differentiate between the OSI and TCP/IP reference models.

Ans.

- ◇ The OSI model consists of seven layers ranging from the Application layer to the Physical layer, whereas the TCP/IP model has four layers: Application, Transport, Internet, and Link.
- ◇ The OSI model was designed to clearly distinguish between services, interfaces, and protocols, providing conceptual clarity. In contrast, the original TCP/IP model did not emphasize these distinctions, focusing instead on practical implementation.
- ◇ OSI offers greater flexibility, allowing protocols to be replaced easily as long as the service definitions are maintained. In TCP/IP, protocols are more tightly integrated, making modifications more challenging without affecting the overall structure.
- ◇ The OSI model aligns well with object-oriented thinking, where services are like methods, interfaces are like parameters/results, and protocols resemble internal code. TCP/IP, however, was not object-oriented by design, though later interpretations tried to align it with that perspective.
- ◇ Regarding connection orientation, the OSI model's network layer supports both connectionless and connection-oriented services, while its transport layer supports only connection-oriented communication. On the other hand, the TCP/IP model's network layer supports only connectionless service through IP, but its transport layer supports both connection-oriented (TCP) and connectionless (UDP) communication.

- ◇ The OSI model did not originally focus on internetworking, as it assumed national OSI-based networks. In contrast, internetworking was a core principle in the design of the TCP/IP model, which aimed to connect heterogeneous networks globally. Lastly, the OSI model is protocol-independent, emphasizing structure and services without defining specific protocols, while the TCP/IP model is protocol-specific, built around foundational protocols like IP, TCP, and UDP.

13. Write a note on the roles of services, interfaces, and protocols in layered network models:

Ans.

In layered models, services are functionalities that a layer provides to the layer above it. Interfaces define how the upper layer accesses the services of the lower layer, and are implemented using service primitives. Protocols are rules and conventions used by corresponding layers in different systems to communicate. For example, the transport layer offers reliable delivery service, accessed through an interface, and uses TCP protocol to manage communication. These concepts ensure abstraction, modularity, and effective communication across network layers.
